

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

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# One-variable data: Distributions and measures of center and spread

# 03

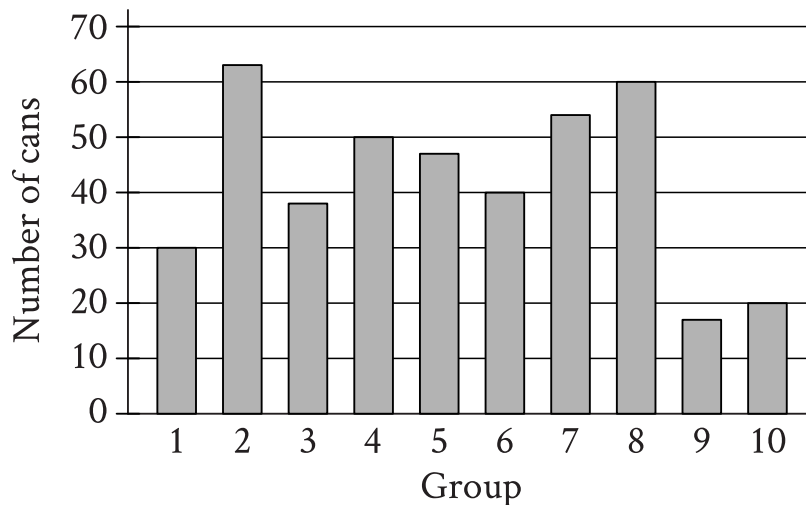
10 questions · ~15 min



Q1

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS



- The data for the 10 categories are as follows:

- Group 1: 30
- Group 2: 63
- Group 3: 38
- Group 4: 50
- Group 5: 47
- Group 6: 40
- Group 7: 54
- Group 8: 60
- Group 9: 17
- Group 10: 20

The bar graph shows the distribution of 419 cans collected by 10 different groups for a food drive. How many cans were collected by group 6?

STUDENT-PRODUCED RESPONSE

|                      |                      |                      |                      |
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|---|---|---|---|
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The high temperature, in degrees Fahrenheit ( $^{\circ}\text{F}$ ), in a certain city was recorded for each of 5 days. The data are shown below.

| Day                                     | 1  | 2  | 3  | 4  | 5  |
|-----------------------------------------|----|----|----|----|----|
| High temperature ( $^{\circ}\text{F}$ ) | 81 | 80 | 81 | 81 | 82 |

Over this 5-day period, which of the following is NOT equal to  $81^{\circ}\text{F}$ ?

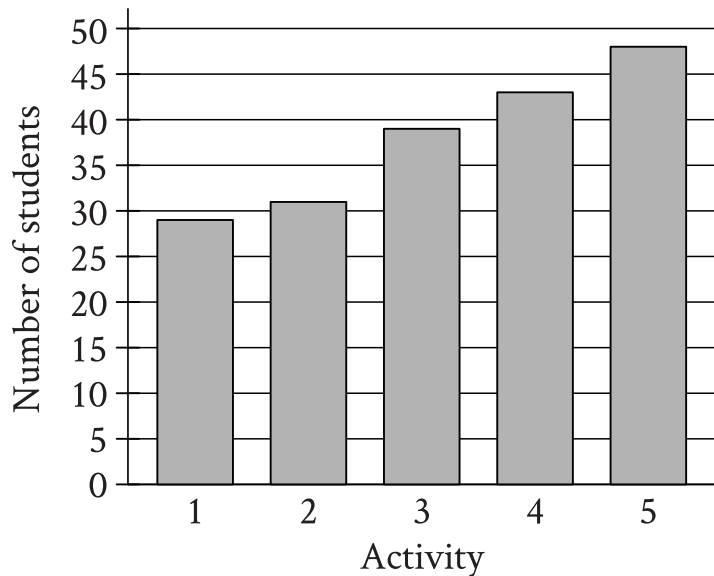
- A. Median of the high temperatures
- B. Mean of the high temperatures
- C. Mode of the high temperatures
- D. Range of the high temperatures

Data set A: 72,73,73,76,76

Data set B: 61,64,74,85,x

Data set A and data set B each contain 5 numbers. If the mean of data set A is equal to the mean of data set B, what is the value of x ?

- A. 77
- B. 85
- C. 86
- D. 95

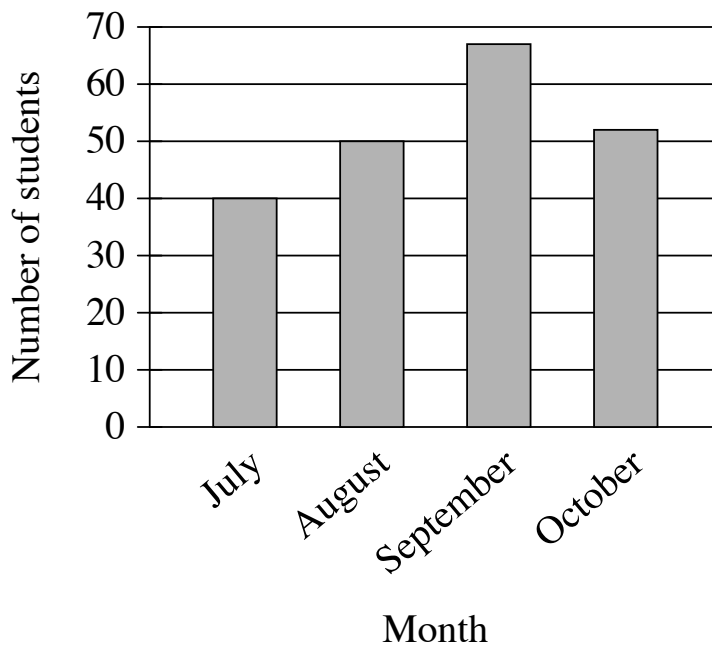


- The data for the 5 categories are as follows:
  - 1: More than halfway above 25 students
  - 2: Less than halfway above 30 students
  - 3: More than halfway above 35 students
  - 4: About halfway above 40 students
  - 5: About halfway above 45 students

A group of students voted on five after-school activities. The bar graph shows the number of students who voted for each of the five activities. How many students chose activity 3?

- A. 25
- B. 39
- C. 48
- D. 50

The bar graph shows the distribution of the number of students from one school who were born in one of four months.

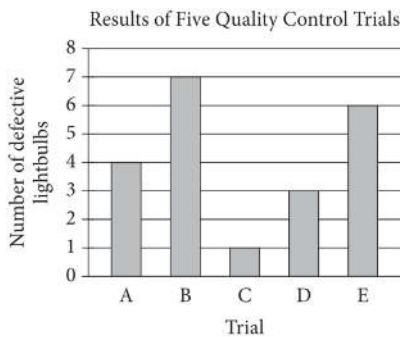


- The Number of students data for the 4 bars are as follows:
  - July: 40
  - August: 50
  - September: 67
  - October: 52

How many more students were born in August than were born in July?

- A. 90
- B. 50
- C. 40

D. 10



For quality control, a company that manufactures lightbulbs conducted five different trials. In each trial, 500 different lightbulbs were tested. The bar graph above shows the number of defective lightbulbs found in each trial. What is the mean number of defective lightbulbs for the five trials?

- A. 4.0
- B. 4.2
- C. 4.6
- D. 5.0

6, 6, 8, 8, 8, 10, 21

Which of the following lists represents a data set that has the same median as the data set shown?

- A. 4, 6, 6, 6, 8, 8
- B. 6, 6, 8, 8, 10, 10
- C. 6, 8, 10, 10, 10, 12
- D. 8, 8, 10, 10, 21, 21

Five *Eretmochelys imbricata*, a type of sea turtle, each have a nest. The table shows an original data set of the number of eggs that each turtle laid in its nest.

| Nest | Number of eggs |
|------|----------------|
| A    | 149            |
| B    | 144            |
| C    | 148            |
| D    | 136            |
| E    | 139            |

A sixth nest with 121 eggs is added to create a new data set. Which of the following correctly compares the means of the two data sets?

- A. The mean of the original data set is greater than the mean of the new data set.
- B. The mean of the original data set is less than the mean of the new data set.
- C. The means of both data sets are equal.
- D. There is not enough information to compare the means.

A study was conducted on the production rates for a company that produces tractor wheels. The table below shows the number of wheels made during 11 consecutive one-hour production periods.

| One-hour period | Number of wheels made |
|-----------------|-----------------------|
| A               | 24                    |
| B               | 24                    |
| C               | 21                    |
| D               | 21                    |
| E               | 21                    |
| F               | 19                    |
| G               | 24                    |
| H               | 24                    |
| I               | 19                    |
| J               | 22                    |
| K               | 23                    |

What is the range of the number of wheels made for the 11 one-hour periods?

A. 5.5

**B.** 5.0

**C.** 4.5

**D.** 4.0

For a school fund-raiser, 10 students sold a total of 90 boxes of cookies. Which of the following can be calculated from this information?

- A. The average number of boxes sold per student
- B. The median number of boxes sold per student
- C. The greatest number of boxes sold by one student
- D. The least number of boxes sold by one student